Topic: Separation of variables

Question: Using the separation of variables, rewrite the partial differential equation as a pair of ordinary differential equations.

$$2\frac{\partial^2 u}{\partial x^2} - 3\frac{\partial u}{\partial y} = 0$$

Answer choices:

A $v''(x) + \lambda v(x) = 0$ and $3w'(y) + 2\lambda w(y) = 0$

B $3v''(x) + \lambda v(x) = 0$ and $2w'(y) + \lambda w(y) = 0$

C $v''(x) - 2\lambda v(x) = 0$ and $w'(y) - 3\lambda w(y) = 0$

D $v''(x) - 3\lambda v(x) = 0$ and $w'(y) - 2\lambda w(y) = 0$

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Solution: A

Starting with the product solution and its partial derivatives,

$$u(x,y) = v(x)w(y)$$

$$\frac{\partial u}{\partial x} = v'(x)w(y) \qquad \frac{\partial u}{\partial y} = v(x)w'(y)$$

$$\frac{\partial^2 u}{\partial x^2} = v''(x)w(y)$$

we'll plug these into the partial differential equation.

$$2v''(x)w(y) - 3v(x)w'(y) = 0$$

$$2\frac{d^2v}{dx^2}w(y) = 3v(x)\frac{dw}{dy}$$

Separate variables then set the equation equal to the separation constant.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = \left(\frac{3}{2w(y)}\right)\frac{dw}{dy}$$

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = \left(\frac{3}{2w(y)}\right)\frac{dw}{dy} = -\lambda$$

Break this into two equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda \qquad \left(\frac{3}{2w(y)}\right)\frac{dw}{dy} = -\lambda$$

$$\frac{d^2v}{dx^2} = -\lambda v(x) \qquad 3\frac{dw}{dy} = -2\lambda w(y)$$

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$$v''(x) + \lambda v(x) = 0$$

$$3w'(y) + 2\lambda w(y) = 0$$

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Topic: Separation of variables

Question: Solve the partial differential equation by separating variables, and assuming that the separation constant is positive.

$$\frac{\partial^2 u}{\partial x^2} - 9\frac{\partial^2 u}{\partial y^2} = 0$$

Answer choices:

A $u(x, y) = (c_1 \cos 3x + c_2 \sin 3x)(c_3 \cos y + c_4 \sin y)$

B $u(x, y) = (c_1 e^{\sqrt{-\lambda}x} + c_2 e^{-\sqrt{-\lambda}x})(c_3 e^{3\sqrt{-\lambda}y} + c_4 e^{-3\sqrt{-\lambda}y})$

C $u(x, y) = (c_1 e^{\sqrt{-\lambda}x} + c_2 e^{-\sqrt{-\lambda}x})(c_3 e^{\frac{\sqrt{-\lambda}}{3}y} + c_4 e^{-\frac{\sqrt{-\lambda}}{3}y})$

D $u(x, y) = (c_1 \cos 3\sqrt{-\lambda}x + c_2 \sin 3\sqrt{-\lambda}x)(c_3 \cos \sqrt{-\lambda}y + c_4 \sin \sqrt{-\lambda}y)$

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Solution: C

Starting with the product solution and its partial derivatives,

$$u(x,y) = v(x)w(y)$$

$$\frac{\partial u}{\partial x} = v'(x)w(y) \qquad \frac{\partial u}{\partial y} = v(x)w'(y)$$

$$\frac{\partial^2 u}{\partial x^2} = v''(x)w(y) \qquad \frac{\partial^2 u}{\partial y^2} = v(x)w''(y)$$

we'll plug these into the partial differential equation.

$$v''(x)w(y) - 9v(x)w''(y) = 0$$

$$v''(x)w(y) = 9v(x)w''(y)$$

$$\frac{d^2v}{dx^2}w(y) = 9v(x)\frac{d^2w}{dy^2}$$

Separate variables then set the equation equal to the separation constant.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = \left(\frac{9}{w(y)}\right)\frac{d^2w}{dy^2}$$

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = \left(\frac{9}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

Because we're using the separation constant $-\lambda$, this value will only be positive when $\lambda < 0$. We'll break this into two equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda \qquad \left(\frac{9}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

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$$\frac{d^2v}{dx^2} = -\lambda v(x) \qquad \frac{d^2w}{dy^2} = -\frac{1}{9}\lambda w(y)$$

$$v'' + \lambda v = 0 \qquad w'' + \frac{1}{9}\lambda w = 0$$

The associated characteristic equations are

$$r^2 = -\lambda \qquad r^2 = -\frac{\lambda}{9}$$

$$r = \pm\sqrt{-\lambda} \qquad r = \pm\frac{\sqrt{-\lambda}}{3}$$

Since $\lambda < 0$, we get the general solutions

$$v(x) = c_1 e^{\sqrt{-\lambda}x} + c_2 e^{-\sqrt{-\lambda}x} \qquad w(y) = c_3 e^{\frac{\sqrt{-\lambda}}{3}y} + c_4 e^{-\frac{\sqrt{-\lambda}}{3}y}$$

So the product solution gives us the general solution to the partial differential equation.

$$u(x,y) = (c_1 e^{\sqrt{-\lambda}x} + c_2 e^{-\sqrt{-\lambda}x})(c_3 e^{\frac{\sqrt{-\lambda}}{3}y} + c_4 e^{-\frac{\sqrt{-\lambda}}{3}y})$$

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Topic: Separation of variables

Question: Solve partial differential equation by separating variables, and assuming that the separation constant is negative.

$$\frac{\partial^2 u}{\partial t^2} - 4\frac{\partial u}{\partial x} = 0$$

Answer choices:

A $u(x,t) = c_1 \cos\sqrt{\lambda}t + c_2 \sin\sqrt{\lambda}t$

B $u(x,t) = c_3 e^{-\lambda x}(c_1 \cos 2\sqrt{\lambda}t + c_2 \sin 2\sqrt{\lambda}t)$

C $u(x,t) = c_3 e^{\sqrt{\lambda}x}\left(c_1 \cos\left(\frac{\lambda}{4}t\right) + c_2 \sin\left(\frac{\lambda}{4}t\right)\right)$

D $u(x,t) = ce^{-\frac{\lambda}{4}x}e^{\sqrt{\lambda}t}$

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Solution: B

Starting with the product solution and its partial derivatives,

$$u(x,t) = v(x)w(t)$$

$$\frac{\partial u}{\partial x} = v'(x)w(t) \qquad \frac{\partial u}{\partial t} = v(x)w'(t)$$

$$\frac{\partial^2 u}{\partial t^2} = v(x)w''(t)$$

we'll plug these into the partial differential equation.

$$v(x)w''(t) - 4v'(x)w(t) = 0$$

$$v(x)w''(t) = 4v'(x)w(t)$$

$$v(x)\frac{d^2w}{dt^2} = 4w(t)\frac{dv}{dx}$$

Separate variables then set the equation equal to the separation constant.

$$\left(\frac{1}{4w(t)}\right)\frac{d^2w}{dt^2} = \left(\frac{1}{v(x)}\right)\frac{dv}{dx}$$

$$\left(\frac{1}{4w(t)}\right)\frac{d^2w}{dt^2} = \left(\frac{1}{v(x)}\right)\frac{dv}{dx} = -\lambda$$

Because we're using the separation constant $-\lambda$, this value will only be negative when $\lambda > 0$. We'll break this into two equations.

$$\left(\frac{1}{4w(t)}\right)\frac{d^2w}{dt^2} = -\lambda \qquad \left(\frac{1}{v(x)}\right)\frac{dv}{dx} = -\lambda$$

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$$\frac{d^2w}{dt^2} + 4\lambda w(t) = 0$$

To solve the first equation, we'll write its associated characteristic equation.

$$r^2 + 4\lambda = 0$$

$$r = \pm 2\sqrt{-\lambda}$$

Since $\lambda > 0$, we get the general solution

$$w(t) = c_1 \cos 2\sqrt{\lambda}t + c_2 \sin 2\sqrt{\lambda}t$$

To solve the second equation, we'll solve it as a separable differential equation.

$$\int \frac{1}{v}\, dv = -\int \lambda\, dx$$

$$\ln|v| = -\lambda x + C$$

$$|v| = c_3 e^{-\lambda x}$$

$$v(x) = c_3 e^{-\lambda x}$$

So the product solution gives us the general solution to the partial differential equation.

$$u(x,t) = c_3 e^{-\lambda x}(c_1 \cos 2\sqrt{\lambda}t + c_2 \sin 2\sqrt{\lambda}t)$$

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